# SMART INDIA HACKATHON 2025

## EUREKA - NEURONEXUS

- **Problem Statement ID –** 25226
- **Problem Statement Title-** AI Tool for Early-Stage Dementia Detection
- **Theme-** MedTech / BioTech / HealthTech
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal) -** Eureka

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# NEURO NEXUS

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## Proposed Solution -

**1. NeuroNexus** is an AI-driven cognitive health companion that unites **multimodal analysis**, digital biomarkers, and predictive modeling for **early detection** and accurate classification.

**2. Multi-Modal Data Capture**: Using smartphones, we gather rich datasets through:

- **Active cognitive tasks-** speech, memory, cognitive function.
- **Passive biomarkers-** voice tone, gaze patterns, facial cues.

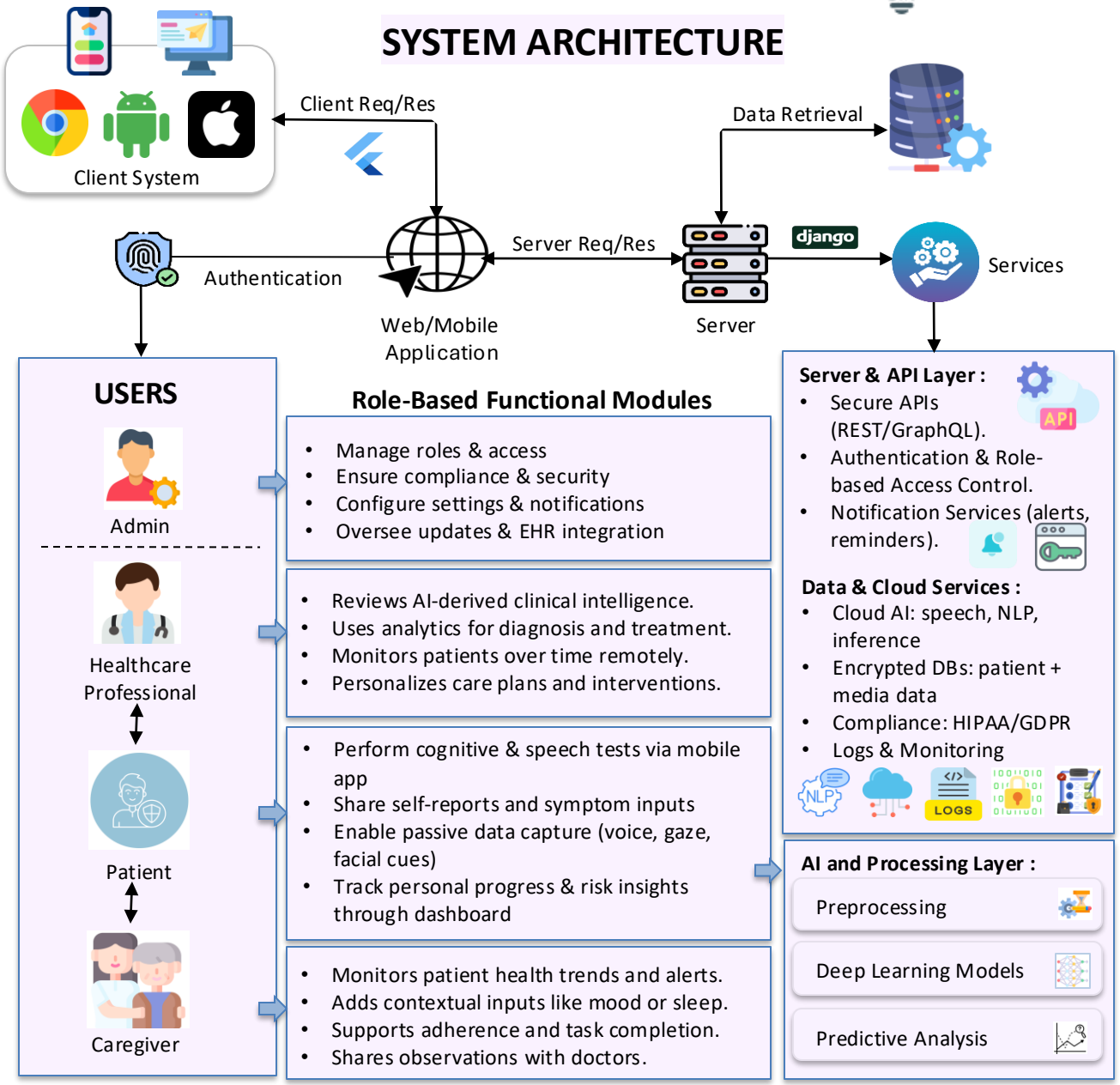
**3. Data Analysis:**

- **Real-Time Analysis** (instant feedback),
- **Deep Learning** (multimodal classification),
- **Predictive Analytics** (forecasting progression and outcomes)

4. Neuro Nexus offers an **elder-friendly**, intuitive **interface** with a built-in **read-aloud assistant** and **multilingual** support, making complex cognitive assessments simple, **accessible**, and engaging.

5. Neuro Nexus turns cognitive assessment into an **engaging**, **game-like experience** challenging memory, attention, language, spatial, and cognitive skills while keeping users **motivated** and **immersed**.

## SYSTEM ARCHITECTURE



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# TECHNICAL APPROACH

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| Feature | Tech Stack | Why it Works |
|---|---|---|
| Cognitive, Speech & Facial Tasks | Flutter, Whisper, MediaPipe,OpenCV | AI analyzes speech, face & emotion |
| Visuospatial & AR Tasks | Flutter, ARCore/ARKit | Measures spatial memory via AR |
| Risk Score & Progress | FastAPI, XGBoost, Firebase Analytics | Generates scores & dashboards |
| Multilingual Support | Flutter, Hugging Face, PyTorch | Supports multiple languages |
| Accessibility & UI | Flutter, Riverpod, Figma | Elder-friendly interface |
| Deployment and Version Control | Docker, Github, App/Play Store | Smooth CI/CD & scalable. |

**Data Collection**
- Cognitive tasks (user tasks)
- Speech & behavior samples (voice and text)
- Face capture (eye gaze, expressions)

**Preprocessing**
- Speech-to-text
- Noise removal & normalization
- Feature extraction (memory, response time etc.)

**AI/ML Analysis**
- NLP models for speech
- ML/DL models for cognitive decline
- Baseline comparison for deviations

**Insights & Actions**
- Risk score & alerts
- Progress tracking
- Dashboard, reports, referrals

***Demo Video Link:*** https://tinyurl.com/3pebtp2c
***Prototype Link:*** https://tinyurl.com/4m2c9k8y

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# FEASIBILITY AND VIABILITY

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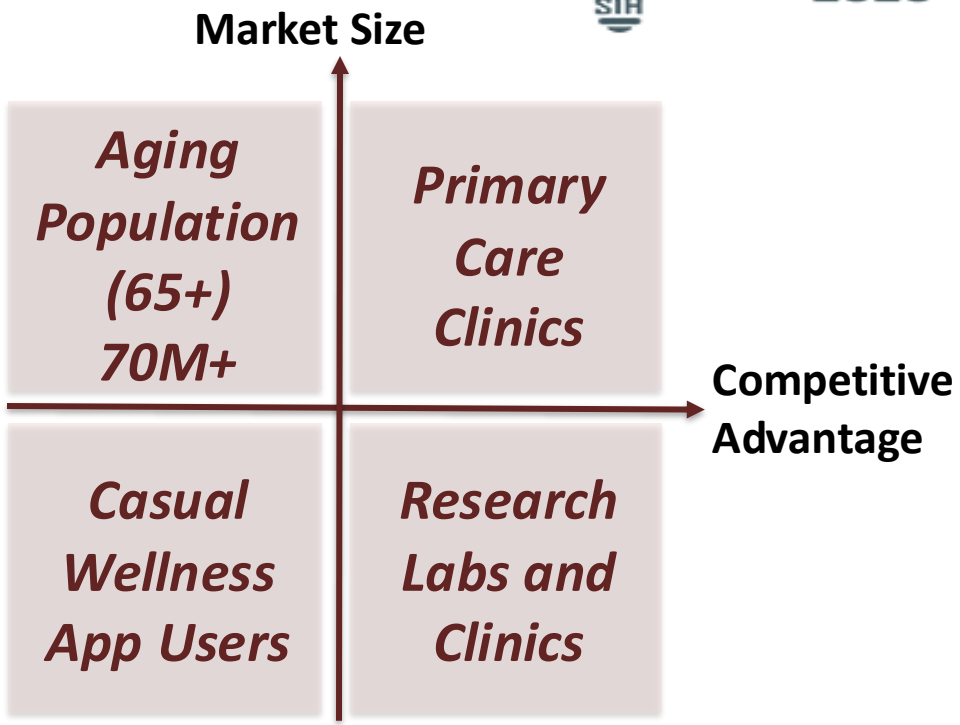
## FEASIBILITY

| Technical | Data | Human | Ethical |
|---|---|---|---|
| ✓ Pretrained AI models<br>✓ Multimodal support<br>✓ Edge & cloud deploy | ✓ Dementia datasets<br>✓ Consent & anonymous<br>✓ Secure storage | ✓ Elder-friendly UI<br>✓ Minimal training<br>✓ Clinician validation | ✓ Regulatory Compliant<br>✓ Data Privacy Laws<br>✓ Bias mitigation |

## VIABILITY

| Clinical | Data | Technical | Social |
|---|---|---|---|
| ✓ Early Detection<br>✓ AI Supports Clinicians<br>✓ Vernacular Access | ✓ Rising Demand<br>✓ Mobile/ Web reach<br>✓ Cost effective | ✓ EHR Integration<br>✓ Cloud/Edge Scalable<br>✓ Model Updates | ✓ Less system burden<br>✓ Aids caregivers<br>✓ Research Data |




## CHALLENGES AND APPROACHES

- **Alignment** → Multimodal sync
- **Noise** → Preprocessing & denoising
- **Language** → Fine-tune + vernacular data
- **Data scarcity** → Synthetic data, transfer learning
- **Overfitting** → Augmentation, validation

# IMPACT AND BENEFITS

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## BENEFITS

### Educational
- Raises symptom awareness.
- Provides digital resources.
- Aids medical training.

### Economical
- Reduces initial and long-term costs.
- Makes early detection affordable.
- Expands access to underserved areas.

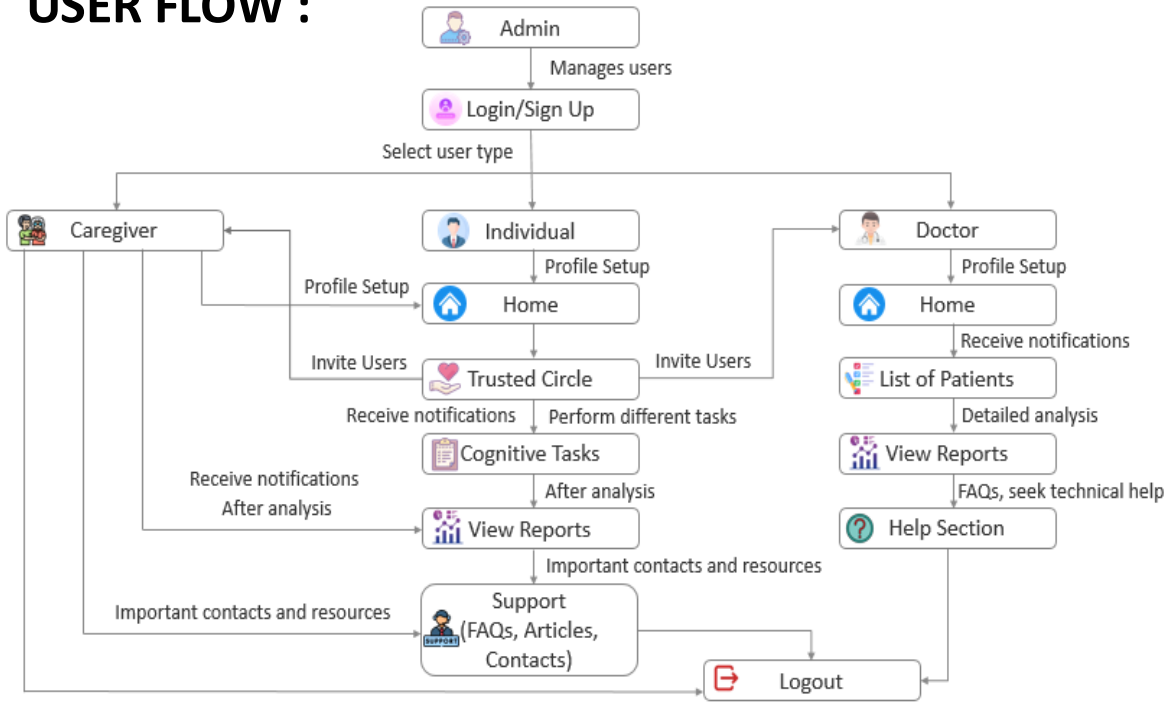
### Social
- Reduces stigma.
- Empowers families early.
- Enables timely support.

### Technical
- Powered by AI analytics.
- Multilingual support.
- Mobile/web accessible.

### Environmental
- Paperless, digital-first approach.
- Reduces travel to clinics, cutting carbon emissions.

## IMPACT :
- Digital dementia screening could reach the 41+ million undiagnosed cases globally.
- Early intervention extends quality-adjusted life years (QALYs) by 0.02-0.03 per person
- Multi-modal data fusion increases diagnostic accuracy compared to single-point assessments
- Predictive healthcare analytics can prevent 25-40% of emergency hospital admissions through early risk detection

## USER FLOW :



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# RESEARCH AND REFERENCES

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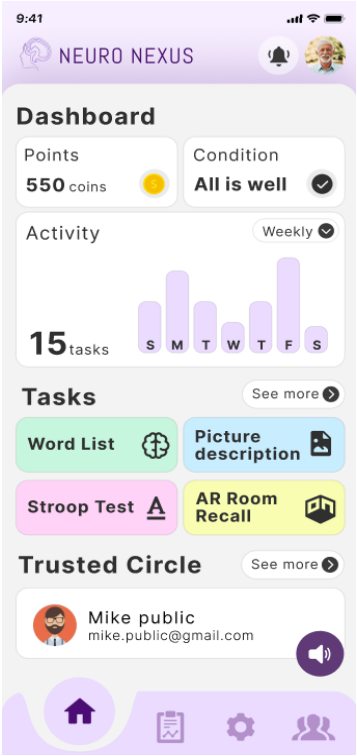
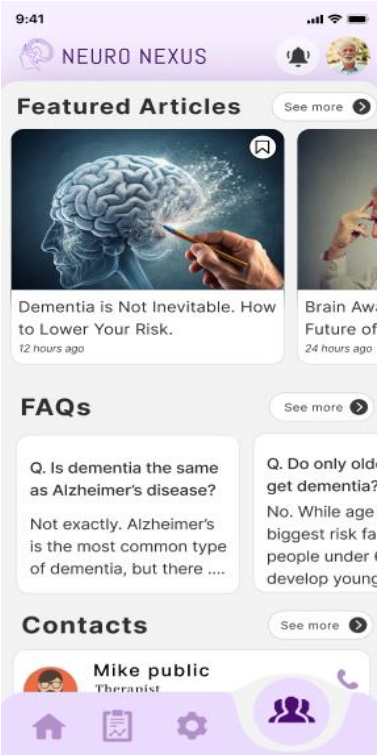
## Research Papers

- Springer Nature - https://tinyurl.com/mvzwcpj2
- Cornell University - https://tinyurl.com/5wrvm265
- Digital Medicine - https://tinyurl.com/2jfcnh2d
- National Library of Medicine - https://tinyurl.com/3b2nsyr9
- Cornell University - https://tinyurl.com/2we5686w
- National Library of Medicine - https://tinyurl.com/3r3cbrnf






## COMPARISON TABLE

| Feature | Neuro Nexus | MindMate | CogState | Altoida |
|---|---|---|---|---|
| Task Reminders | ✓ | ✓ | ✗ | ✗ |
| Multilingual | ✓ | ✗ | ✗ | ✗ |
| Risk Score (AI Based) | ✓ | ✗ | Limited | Limited |
| Facial Analysis | ✓ | ✗ | ✗ | Limited |
| Elder friendly UI | ✓ | Limited | ✗ | ✗ |
| AI Voice Assist | ✓ | Limited | ✗ | ✗ |
| Support (FAQs & Articles) | ✓ | ✗ | ✗ | ✗ |